

Text, Pixels, or Both? Evaluating Input Representations for Multimodal Document QA

Nikhil Reddy Pottanigari Sepideh Kharaghani Saverio Vadicchino
Alejandro Posada Kurt MacDonald Ying Zhang

ServiceNow Canada

{nikhilreddy.pottanigari, sepideh.kharaghani,
saverio.vadicchino, alejandro.posada,
kurt.macdonald, yin.zhang}@servicenow.com

Abstract

Every document QA system begins with a choice that is rarely studied on its own: whether to feed the model page images, extracted text, or both. We isolate this choice, holding the prompt, judge, and scoring pipeline fixed, across four commercial model endpoints, two corpora, and two context regimes (gold evidence pages and the full document). On documents that fit the image budget, page images lead on accuracy at every document length on both corpora, but this advantage carries a growing latency and cost premium: text latency stays roughly flat as documents lengthen while image latency rises steadily. Text and images also fail on different questions, with exactly one representation correct on **19–25%** of items across the reported cells, so neither subsumes the other. Exploiting this complementarity, a lightweight TF-IDF router that reads only the question text gains **2.6 points** over always-text while cutting median latency **30%** relative to always-vision, on a document-disjoint held-out split.

1 Introduction

Before a document QA system can answer a question, a simple preprocessing choice sets how the document reaches the model: as page images, as extracted text, or both (Zhang et al., 2024). A knowledge-base article may encode evidence in a screenshot, diagram, or spatial arrangement that OCR text preserves only partially. Sending every page as an image retains this visual evidence but increases input size and inference latency on long documents. Combining both modalities can be worse than either channel alone, depending on the model.

This choice affects accuracy and latency together. Always sending images for a long report, or text for a screenshot-heavy knowledge base, can hurt answer quality or increase inference time. However, representation is often evaluated together with

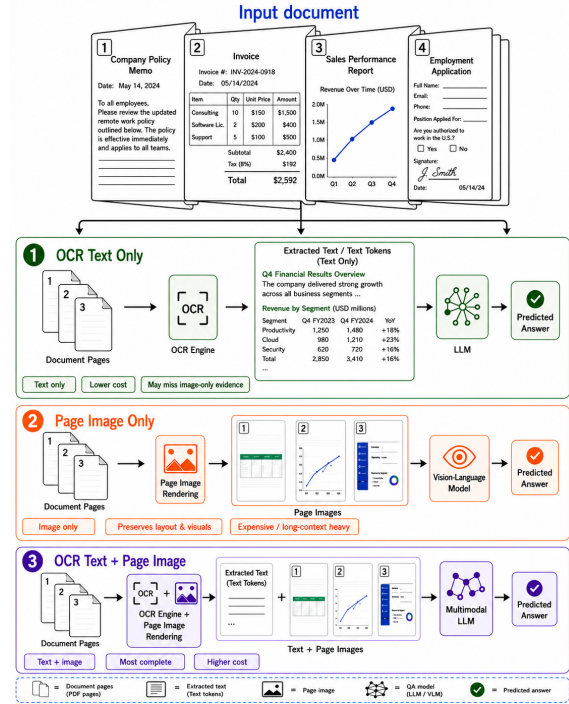


Figure 1: Three document representations use the same QA prompt and judge. **IMAGE** preserves layout and visual content. **TEXT** is compact and uncapped by page count. **TEXT+IMAGE** concatenates OCR text followed by page images at the cost of larger context.

changes to the model, prompt, retrieval method, or evaluator, making its contribution difficult to identify.

We address this by comparing **IMAGE** (page images), **TEXT** (OCR text), and **TEXT+IMAGE** (text then images) across four commercial endpoints from two providers, one public benchmark (Ma et al., 2024b), and one enterprise knowledge base. Each model runs under two retrieval regimes: **ORACLE** (gold evidence pages) and **FULL** (full document). Within each model, corpus, and regime, the prompt, judge, and scoring pipeline are held constant; the representation and its resulting input size are the intended experimental differences.

Table 1: Evaluation corpora. The ≤ 50 -page columns count the documents short enough to fit the vision page cap and their questions (the subset used in Table 3).

Corpus	Docs	≤ 50 -page Docs	Questions	≤ 50 -page Questions	Profile
Enterprise KB	1,447	1,445	5,169	5,163	Short support articles with screenshots, UI states, diagrams, and cross-modal references; median 4pp., max 237pp.; 99.86% of docs ≤ 50 pp.
MMLongBench-Doc	135	101	1,091	798	Visually rich PDFs: reports, papers, manuals; cross-page and unanswerable questions (Ma et al., 2024b); median 28pp., max 468pp. (mean 48.3); 25% of docs > 50 pp.

Our contributions:

- A **controlled representation benchmark** across four models, two corpora, and two retrieval regimes with a fixed prompt and judge.
- A **systematic analysis of how document length interacts with input representation**, showing that **IMAGE** leads on accuracy within the image budget while their latency and cost grow with document length (Section 4.1).
- A **lightweight question-text router** that captures much of the accuracy gap to always-vision while retaining most of text’s latency advantage, evaluated on a document-disjoint held-out split (Section 5).

2 Related Work

Document QA and representation. Document QA has expanded from single-page visual understanding (Mathew et al., 2021) to multi-page reasoning over tables, charts, and cross-page evidence (Van Landeghem et al., 2023; Ma et al., 2024b; Deng et al., 2025). Layout-aware models such as LayoutLMv3 (Huang et al., 2022) and Pix2Struct (Lee et al., 2023) learn joint text–image representations. Berghaus et al. (2025) compare image- and text-based processing for invoice extraction and report model- and dataset-dependent results. Our experiments compare representations under a common prompt and evaluation pipeline across four models, two corpora, and two regimes.

Multimodal document retrieval and benchmarks. Retrieval-augmented generation (Lewis et al., 2020) couples two decisions: which evidence to retrieve and how to represent it. PDF-Triage (Saad-Falcon et al., 2024) routes questions to document structure; M3DocRAG (Cho et al., 2024) and MMDocRAG (Dong et al., 2025) retrieve multimodal document evidence. Our ORACLE/FULL regimes compare gold-page input with

no page retrieval. Long-context position effects are established for text (Liu et al., 2024), and prompt-compression methods reduce text context length (Jiang et al., 2023; Li et al., 2023). We examine how document representation interacts with full-document context and a fixed image budget. A parallel line of work asks whether page images can replace extracted text directly. ColPali (Faysse et al., 2025) and Document Screenshot Embedding (Ma et al., 2024a) embed rendered pages rather than parsed text, and VisRAG (Yu et al., 2025) builds a vision-language RAG pipeline over page images, each reporting that image representations avoid parsing loss on layout-rich documents. Our controlled comparison isolates this same text-versus-image choice at generation time rather than at retrieval time.

Parsing and evaluation. Text quality depends on the parsing pipeline. Neural parsers (Blecher et al., 2024; Nassar et al., 2025) produce structured markup from page images, while direct multimodal processing can retain non-textual evidence that text extraction omits (Shen et al., 2026). Our text condition uses a proprietary pipeline that outperforms Tesseract (Smith, 2007) on three OCR datasets (Appendix Table 8); this comparison does not eliminate extraction error as a possible source of the text–image gap. We use the same LLM judge and prompt in every condition, while recognizing that LLM-based evaluation can itself introduce error (Zheng et al., 2023; Liu et al., 2023; Bavaresco et al., 2025).

3 Setup

Representation strategies. **IMAGE** sends each page as a 300-DPI image; **TEXT** sends page text from a proprietary OCR pipeline whose comparison with Tesseract (Smith, 2007) appears in Appendix Table 8; **TEXT+IMAGE** supplies the OCR text followed by the page images. All three use the

Table 2: Evaluation metrics: accuracy as the primary outcome; pairwise disagreement, and latency

Metric	Definition	Interpretation
Accuracy	$\text{Acc}(s, m, d, r) = Q ^{-1} \sum_q c_q$, with $c_q \in [0, 1]$ the mean judged correctness across prediction and judge runs.	Average judged answer correctness; higher is better.
Pairwise disagreement	$\text{Dis}(s_i, s_j) = Q ^{-1} \sum_q \mathbf{1}[b_{q,i} \neq b_{q,j}]$, where $b_{q,s}$ is the binary correctness label used for paired analysis.	Higher values indicate more complementary correctness outcomes.
Latency	Average seconds per question from run telemetry, by representation and regime.	High latency can make an accurate route impractical.

same QA prompt (Appendix E).

Datasets. Two corpora span short, visually dense articles and longer mixed-format documents (Table 1). The enterprise knowledge base (KB) averages 5.7 pages per document and has 93.6% of its questions labeled as requiring visual or cross-modal evidence. MMLongBench-Doc (Ma et al., 2024b) contains longer PDFs (48.3 pages on average) with single-page, cross-page, and unanswerable questions. The KB is internal, whereas MMLongBench-Doc provides a public comparison corpus (Appendix B).

Models and retrieval regimes. We evaluate four models from two providers: Gemini 3.1 Flash-Lite and Gemini 3.1 Pro (Google via Vertex AI), GPT-5.4 and GPT-5.4-mini (OpenAI via Azure)¹. Each model runs under two regimes that bracket realistic retrieval:

- **ORACLE** supplies only the gold evidence pages. It measures answer generation when annotated evidence pages are provided, without evaluating page retrieval.
- **FULL** sends the full document with no retrieval. Here the model has to both find the relevant pages and extract the answer from a much larger, noisier context.

The gap between regimes measures sensitivity to replacing annotated evidence pages with the full document: a large positive gap indicates that the representation benefits from page selection, whereas a small or negative gap indicates comparable or better performance with full context.

Page cap and document scope. Due to model-provider limits, we cap all requests at **50 page images** and restrict the main analysis to documents of 50 pages or fewer, so that every representation sees

the complete page set. We adopt this restriction specifically to isolate the page-cap truncation effect from any genuine modality effect. Applying the same cap to both providers keeps the **IMAGE** conditions comparable. For completeness, Appendix C reports results over *all* documents, still capped at 50 page images. In that setting, documents longer than 50 pages are truncated under **IMAGE**, while **TEXT** continues to see the full document, and the two effects are shown to be separable.

Scoring and metrics. Models produce free-form answers with a 3,000-token limit, temperature 0, and a fixed seed of 30. For each model, representation, dataset, and regime, we average correctness over **five prediction runs**. Each prediction is then evaluated by a fixed GPT-5.2 judge over **three judging runs**, and we report the mean score. Per-cell run-to-run standard deviations are reported alongside every accuracy and latency value in Tables 3 and 4. Reported $\pm$ values are standard deviations across the five prediction and three judging runs. Because decoding is greedy with a fixed seed, they quantify residual nondeterminism in the hosted endpoints and the judge rather than sampling uncertainty over questions. Table 2 defines the three metrics.

4 Results

We organize results around two questions: how the accuracy–latency trade-off changes with document length, and whether text and images miss the same questions. Table 3 and Figure 2 show that on the ≤ 50 -page subset, image-bearing inputs (**IMAGE**, **TEXT+IMAGE**) lead on accuracy across document lengths on both corpora, while **TEXT** stays consistently lower but at a fraction of the latency: text latency is roughly flat in document length, whereas **IMAGE** and **TEXT+IMAGE** latency grows steadily. The accuracy gap narrows on the

¹Exact API identifiers in Appendix A.

Table 3: Accuracy on the ≤ 50 -page subset (documents short enough that every representation sees the full page set, i.e. no vision-input truncation). Cells show mean $\pm$ std across runs, quantifying endpoint and judge nondeterminism rather than question-level uncertainty; the Avg. row is the arithmetic mean of the four model means per column. Left block: ORACLE (gold evidence pages); right block: FULL (full document).

Model	ORACLE			FULL		
	TEXT	IMAGE	TEXT+IMAGE	TEXT	IMAGE	TEXT+IMAGE
Enterprise KB						
Gemini 3.1 Flash-Lite	0.576 $\pm$ 0.002	0.729 $\pm$ 0.002	0.724 $\pm$ 0.001	0.594 $\pm$ 0.001	0.711 $\pm$ 0.003	0.714 $\pm$ 0.000
Gemini 3.1 Pro	0.618 $\pm$ 0.003	0.741 $\pm$ 0.004	0.736 $\pm$ 0.002	0.634 $\pm$ 0.003	0.728 $\pm$ 0.002	0.728 $\pm$ 0.003
GPT-5.4	0.639 $\pm$ 0.003	0.677 $\pm$ 0.002	0.733 $\pm$ 0.004	0.663 $\pm$ 0.006	0.679 $\pm$ 0.004	0.734 $\pm$ 0.005
GPT-5.4-mini	0.612 $\pm$ 0.002	0.683 $\pm$ 0.003	0.731 $\pm$ 0.002	0.628 $\pm$ 0.003	0.672 $\pm$ 0.003	0.715 $\pm$ 0.005
Avg.	0.611	0.708	0.731	0.630	0.698	0.723
MMLongBench-Doc						
Gemini 3.1 Flash-Lite	0.529 $\pm$ 0.001	0.625 $\pm$ 0.003	0.613 $\pm$ 0.002	0.511 $\pm$ 0.001	0.615 $\pm$ 0.009	0.620 $\pm$ 0.006
Gemini 3.1 Pro	0.641 $\pm$ 0.007	0.711 $\pm$ 0.008	0.691 $\pm$ 0.003	0.628 $\pm$ 0.007	0.690 $\pm$ 0.004	0.623 $\pm$ 0.006
GPT-5.4	0.559 $\pm$ 0.007	0.667 $\pm$ 0.004	0.673 $\pm$ 0.003	0.599 $\pm$ 0.009	0.686 $\pm$ 0.008	0.683 $\pm$ 0.008
GPT-5.4-mini	0.527 $\pm$ 0.009	0.634 $\pm$ 0.005	0.643 $\pm$ 0.007	0.547 $\pm$ 0.005	0.633 $\pm$ 0.004	0.642 $\pm$ 0.007
Avg.	0.564	0.659	0.655	0.571	0.656	0.642

Table 4: Mean latency (seconds/question, $\pm$ std across runs) on the ≤ 50 -page subset, where no representation is truncated by the vision page cap. Left block: ORACLE (gold evidence pages); right block: FULL (full document). The Avg. row is the arithmetic mean of the four model means per column. Lower is better.

Model	ORACLE			FULL		
	TEXT	IMAGE	TEXT+IMAGE	TEXT	IMAGE	TEXT+IMAGE
<i>Enterprise KB</i>						
Gemini 3.1 Flash-Lite	1.29 $\pm$ 0.00	2.42 $\pm$ 0.38	2.52 $\pm$ 0.31	1.40 $\pm$ 0.01	3.26 $\pm$ 0.60	3.20 $\pm$ 0.32
Gemini 3.1 Pro	5.57 $\pm$ 0.25	5.69 $\pm$ 0.36	5.36 $\pm$ 0.37	6.21 $\pm$ 0.25	5.68 $\pm$ 0.15	5.81 $\pm$ 0.22
GPT-5.4	3.46 $\pm$ 0.29	4.97 $\pm$ 0.42	4.48 $\pm$ 0.42	3.42 $\pm$ 0.18	5.43 $\pm$ 1.07	5.71 $\pm$ 0.43
GPT-5.4-mini	1.68 $\pm$ 0.20	2.55 $\pm$ 0.15	2.42 $\pm$ 0.15	1.78 $\pm$ 0.42	3.36 $\pm$ 0.13	3.26 $\pm$ 0.16
Avg.	3.00	3.91	3.70	3.20	4.43	4.50
<i>MMLongBench-Doc</i>						
Gemini 3.1 Flash-Lite	1.72 $\pm$ 0.09	3.59 $\pm$ 0.44	4.06 $\pm$ 0.16	2.10 $\pm$ 0.06	5.88 $\pm$ 0.52	6.93 $\pm$ 0.28
Gemini 3.1 Pro	6.10 $\pm$ 0.22	6.88 $\pm$ 0.69	6.80 $\pm$ 0.72	7.57 $\pm$ 0.31	8.40 $\pm$ 0.50	10.12 $\pm$ 2.31
GPT-5.4	4.21 $\pm$ 0.32	8.03 $\pm$ 0.75	7.83 $\pm$ 0.67	4.78 $\pm$ 0.24	13.37 $\pm$ 0.61	13.36 $\pm$ 0.51
GPT-5.4-mini	2.00 $\pm$ 0.23	5.05 $\pm$ 0.36	5.03 $\pm$ 0.31	2.22 $\pm$ 0.27	10.20 $\pm$ 0.31	10.09 $\pm$ 0.36
Avg.	3.51	5.89	5.93	4.17	9.46	10.13

shortest and longest KB buckets but does not close. Section 5 then exploits the resulting complementarity with a lightweight router.

4.1 Accuracy-latency trade-off across representations

Page images retain layout and non-textual content that extracted text may omit (Shen et al., 2026), but any accuracy gain must be weighed against latency. We examine this trade-off under two regimes, on the ≤ 50 -page subset where every representation sees the complete page set.

Oracle: image inputs lead with gold pages. Under ORACLE, either **IMAGE** or **TEXT+IMAGE** is best for every model on both corpora, with gains over **TEXT** of up to 14.8 percentage points on the KB and 11.6 points on MMLongBench-Doc (Ta-

ble 3). This is consistent with image inputs retaining evidence omitted by the text pipeline, but it does not identify whether the difference arises from visual evidence, OCR errors, or model-specific processing. The gain comes at a latency cost: on the KB, image input raises average latency from 3.0 s (**TEXT**) to 3.9 s (**IMAGE**) and 3.7 s (**TEXT+IMAGE**), and on MMLongBench-Doc from 3.5 s to 5.9 s (Table 4). The larger ORACLE latency gap on MMLongBench-Doc is consistent with its questions citing more evidence pages per query (1.88 vs. 1.05 on the KB (Appendix B)), so each **IMAGE** or **TEXT+IMAGE** request carries more page images.

Full: image accuracy leads, but the latency gap widens. Under FULL, image-bearing inputs still lead on accuracy on both corpora: on the

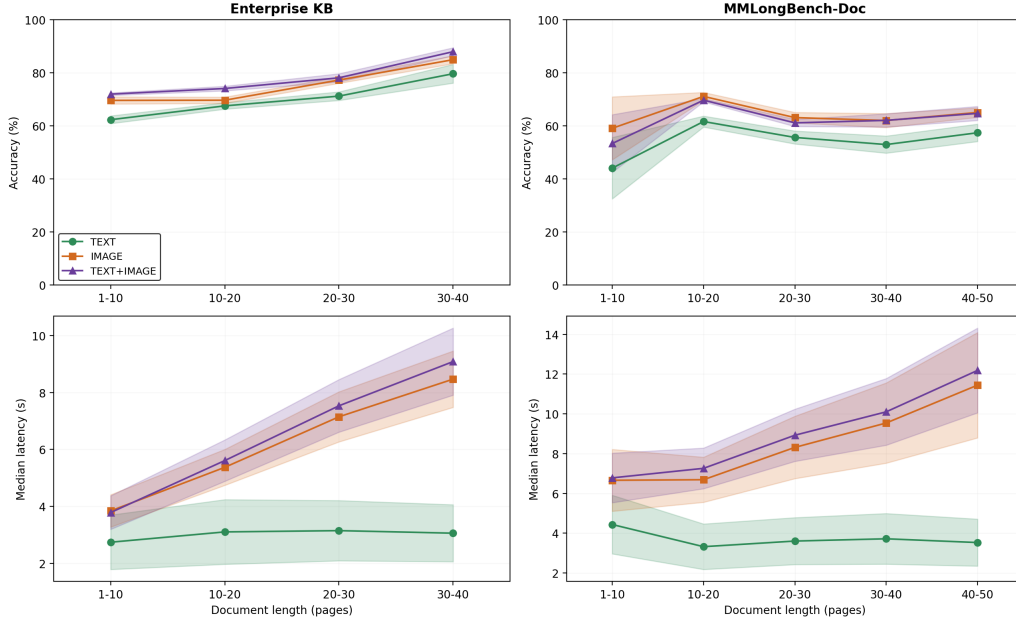


Figure 2: Accuracy (top) and median latency (bottom) versus document-length bucket on the ≤ 50 -page subset under FULL retrieval, aggregated across all four models. The Enterprise KB has no 40–50 page bucket, as its documents are shorter (median 4 pages). Bands show $\pm$ std across models.

KB, **TEXT+IMAGE** and **IMAGE** average 0.72 and 0.70 versus 0.63 for **TEXT**; on MMLongBench-Doc, **IMAGE** averages 0.66 versus 0.57 for **TEXT** (Table 3). Figure 2 shows **TEXT** latency staying roughly flat in document length (~ 3 s on the KB, ~ 3 – 4 s on MMLongBench-Doc), while **IMAGE** and **TEXT+IMAGE** latency rise steadily, reaching ~ 9 s on the longest KB bucket and ~ 12 s on the longest MMLongBench-Doc bucket. Accuracy, by contrast, remains image-favored in every length bucket within the subset. The bucket-level view is included to show that the accuracy ordering does not reverse within the un-truncated range; the reversal reported in Appendix C appears only once the page cap binds.

Does combining both modalities help?

Whether **TEXT+IMAGE** helps depends on the endpoint and corpus. GPT-5.4 and GPT-5.4-mini gain roughly 5 pp (5.6 and 4.8, respectively) from **TEXT+IMAGE** over **IMAGE** on the KB under ORACLE, and their **IMAGE**-vs-**TEXT+IMAGE** disagreement is the highest among the KB cells (13–14%, Table 6), showing that the additional text changes some correctness outcomes. Gemini Pro’s **IMAGE** and **TEXT+IMAGE** outcomes differ less often (~ 7 – 8% on the KB). On MMLongBench-Doc/FULL, Gemini Pro’s **TEXT+IMAGE** falls behind its **IMAGE** result and roughly ties its **TEXT** result (0.623 vs. 0.690 and 0.628), while for the

Table 5: Average cost (USD per 100 questions) and latency (seconds per question) per representation strategy, averaged across the four models and two datasets. Full per-model, per-dataset latency breakdowns are in Table 4.

Strategy	Cost (\$/100q)	Latency (s/q)
ORACLE		
TEXT	0.61	3.25
IMAGE	0.91	4.90
TEXT+IMAGE	1.29	4.81
FULL		
TEXT	1.48	3.69
IMAGE	2.78	6.95
TEXT+IMAGE	3.51	7.31

GPT endpoints **TEXT+IMAGE** is the most accurate condition on the KB. **TEXT+IMAGE** provides no consistent latency benefit and is the slowest condition on the longer corpus (Table 4).

Finding 1. Document length governs the *cost* of the trade-off, not its direction. Within the ≤ 50 -page subset, image-bearing inputs (**IMAGE** or **TEXT+IMAGE**) lead on accuracy on both corpora, with no length bucket showing a text advantage; what grows with length is their latency, which rises steadily while text latency stays roughly flat.

The image advantage carries a cost and latency premium. Table 5 summarizes cost and latency across the four models and two corpora. Within each regime, **TEXT** is consistently the

Table 6: Pairwise disagreement rate (%) on the ≤ 50 -page subset: share of questions where exactly one format is correct (i.e., the sum of text-only-correct and vision-only-correct questions). A rate of 10% in the **TEXT/IMAGE** column means that on 10% of questions one format answered correctly and the other did not. Coral $\geq 20\%$; mint $\leq 10\%$.

Model	Enterprise KB				MMLongBench-Doc			
	TEXT/IMAGE		IMAGE/TEXT+IMAGE		TEXT/IMAGE		IMAGE/TEXT+IMAGE	
	ORACLE	FULL	ORACLE	FULL	ORACLE	FULL	ORACLE	FULL
Gemini 3.1 Flash-Lite	24.7	22.2	10.7	10.4	22.8	23.2	12.3	13.5
Gemini 3.1 Pro	20.4	19.2	6.7	8.4	19.3	21.2	10.4	16.0
GPT-5.4	22.9	19.7	13.5	12.7	19.7	20.2	8.9	7.9
GPT-5.4-mini	24.6	22.6	14.0	14.2	23.4	20.6	9.6	9.5

cheapest and fastest condition: under ORACLE, moving from **TEXT** to **IMAGE** raises cost $1.5\times$ (0.61 to 0.91 \$/100q) and latency from 3.3 to 4.9 s/q, and **TEXT+IMAGE** is more expensive still (1.29 \$/100q).

The premium widens under FULL, where the full document is processed: **IMAGE** and **TEXT+IMAGE** cost $1.9\times$ and $2.4\times$ as much as **TEXT**, respectively, while their latency is roughly twice that of text (7.0 and 7.3 s/q vs. 3.7 s/q). Since **IMAGE** or **TEXT+IMAGE** also lead on accuracy (Section 4.1), no representation dominates on both axes. This trade-off motivates per-question routing, using image input only when it is likely to improve the answer.

4.2 Text and images answer different questions

TEXT and **IMAGE** give different correctness outcomes on approximately **19–25%** of questions across the reported corpus, model, and regime cells (Table 6). This disagreement decomposes into a text-only-correct slice and a larger vision-only-correct slice, reflecting vision’s higher overall accuracy. On the short KB articles, ORACLE and FULL disagreement rates are nearly identical because the full document is already close to the evidence pages. On MMLongBench-Doc the two regimes diverge somewhat more, consistent with extra context amplifying representation differences. Because both exclusive-correct slices are present in every cell, an oracle that picks the correct format per question would beat either fixed policy.

Finding 2. Text and images answer different questions: 19–25% **TEXT/IMAGE** disagreement across the reported cells. Neither format subsumes the other; this establishes headroom for per-question selection.

5 Per-Question Representation Routing

Text and images each answer questions the other misses (Finding 2). This complementarity raises a concrete question: if a system could decide *per question* whether to send text or images, it could capture the accuracy of images where they help while keeping text’s lower latency and cost elsewhere. We test whether this is achievable using only the question text, as a proof of concept that quantifies the potential benefit rather than a production-ready system.

Router design. We frame routing as binary classification. For each (question, model) pair under FULL, the label is $y = 1$ if **IMAGE** is correct and **TEXT** is incorrect, and $y = 0$ otherwise. When both representations are correct we default to **TEXT** because it is faster in aggregate; when both are incorrect we also assign **TEXT**, treating the example as a non-vision-win case. We exclude **TEXT+IMAGE** from the router: although often competitive, it adds a second input channel and closely tracks **IMAGE** in many settings (Table 3), so routing between **TEXT** and **IMAGE** targets the primary accuracy–latency trade-off.

Labels are obtained by running both representations on every question. This makes the router a proof of concept: it measures how much a per-question policy *could* gain if it selected correctly, not the performance of a deployable classifier. In production, a similar router would be trained on a smaller labeled sample and applied to new questions. In our training set only 13.2% of examples are positive, so **IMAGE** uniquely beats **TEXT** on a small fraction of questions.

We represent each question with TF-IDF features and train a logistic regression classifier with balanced class weights, using an approximately 80/20 document-level split stratified by dataset and

Table 7: Held-out question-text router results under FULL, pooling all four models. The test partition is the 20% split described in the text. *Routing accuracy* = accuracy when every question uses the modality the strategy selects (cf. per-model accuracy in Table 3). *Implied med. latency* = median wall-clock seconds across the strategy’s per-question routing decisions (cf. per-model latency in Table 4). Gains are computed from unrounded scores. **Bold** = best per column within each dataset.

Strategy	Routing acc.	Gain vs text	Lat.
Overall			
Always-text	63.6%	+0.0 pp	2.5 s
Always-vision	67.7%	+4.1 pp	4.4 s
TF-IDF router	66.2%	+2.6 pp	3.1 s
Enterprise KB			
Always-text	65.0%	+0.0 pp	2.4 s
Always-vision	69.2%	+4.2 pp	4.1 s
TF-IDF router	67.8%	+2.7 pp	2.9 s
MMLongBench-Doc			
Always-text	53.9%	+0.0 pp	3.2 s
Always-vision	58.0%	+4.1 pp	8.7 s
TF-IDF router	55.9%	+2.0 pp	5.6 s

model. All questions from the same document stay in one partition, ensuring zero document leakage and a more reliable estimate of generalization to unseen documents.

All router numbers in Table 7 are computed on this 20% held-out test partition over the *full corpus* (all documents, image budget still capped at 50 pages) and pooled across the four models, so they are micro-averages over pooled questions rather than the per-model, ≤ 50 -page macro-averages in Table 3; the two are not directly comparable. Always-vision is correspondingly lower on MMLongBench-Doc here, because the held-out set includes documents beyond the 50-page cap where image inputs are truncated (Appendix C).

Accuracy. On the document-disjoint split, always-vision remains the strongest fixed policy, reaching 69.2% on KB and 58.0% on MMLongBench-Doc (Table 7). The router does not beat it, but closes much of the gap over always-text, increasing accuracy by **2.7 pp** on KB (65.0 $\rightarrow$ 67.8%) and **2.0 pp** on MMLongBench-Doc (53.9 $\rightarrow$ 55.9%) while routing 26.8% of examples to vision. Overall, the router reaches 66.2%, gaining **2.6 pp** over always-text while remaining 1.5 pp below always-vision.

Latency. Since the router keeps most questions on the faster text path, median latency drops sub-

stantially relative to always-vision: from 4.4 s to 3.1 s overall, a savings of approximately **30%**, while routing only 26.8% of examples to vision.

Finding 3. On the held-out split with no document overlap, a lightweight TF-IDF router gains 2.0–2.7 pp over always-text while cutting median latency by 30% relative to always-vision. Because training labels require running both representations on every question, this is a proof-of-concept estimate of achievable headroom, not a deployable classifier.

6 Discussion

Document length as the organizing variable.

Document length organizes the accuracy–latency trade-off, but not by changing which representation is most accurate. On the ≤ 50 -page subset, **IMAGE** or **TEXT+IMAGE** leads on accuracy across all document lengths on both corpora; what scales with length is their cost and latency, which grow steadily while text stays roughly flat. The apparent accuracy crossover in favor of text on long documents is a consequence of the fixed 50-image cap truncating image inputs, and appears only in the full-corpus results (Appendix C); it is a property of the current deployment budget, not of the modality. Complementary errors provide a separate motivation for routing: text and images disagree on roughly a fifth to a quarter of questions in every cell, so neither representation subsumes the other. The current router, however, is evaluated only under FULL, and we treat it as a proof of concept rather than a tuned system.

Multimodal fusion is model-dependent, not additive.

Sending both text and images does not uniformly help. The GPT endpoints gain approximately 5 pp over image-only input on the KB under ORACLE, whereas the Gemini endpoints show smaller or less consistent changes (Section 4.1). These outcome differences do not reveal the models’ internal fusion mechanisms, but they show that the preferred representation depends on the endpoint as well as the document setting.

Implications for RAG pipelines. The results motivate evaluating both text and image indices rather than assuming that one representation is uniformly best. The TF-IDF router is a proof of concept; a stronger evaluation would use realistic retrieval, and document features in addition to the question text.

7 Conclusion

We compared text, image, and combined document inputs under a common prompting and evaluation pipeline across four model endpoints, two corpora, and two context regimes. Image-bearing inputs outperform text on both corpora, with the largest gains on the short, visually dense Enterprise KB, while text remains cheaper and faster per question. The two representations also produce complementary correctness outcomes, with approximately 19–25% pairwise disagreement across the reported cells. On a held-out document-level split with no train–test leakage, a lightweight TF-IDF question router recovers 2.0–2.7 percentage points over always-text while reducing median latency by 30% relative to always-vision.

Limitations

First, ORACLE and FULL do not evaluate a realistic top- k retriever. They compare annotated evidence pages against full-document input, so the results bound two extremes and do not establish performance for intermediate retrieval quality.

Second, we report run-to-run variation but not question-level uncertainty. Our design fixes the question set and varies only prediction and judging runs, so the reported standard deviations capture endpoint and judge nondeterminism. Paired significance testing over the per-question correctness scores would sharpen these comparisons.

Third, the **TEXT** condition depends on our OCR pipeline. The pipeline outperforms Tesseract on standard document-OCR benchmarks (Appendix Table 8) but is not state of the art, and may extract text less faithfully than the frontier models themselves would from the same pages. Part of the text–image gap we report may therefore reflect extraction error rather than an intrinsic limitation of the text modality.

Ethics Statement

The Enterprise KB contains internal product-support documentation owned by the authors’ organization and was accessed under existing internal data-use policies. Questions were created and reviewed by internal annotators as part of their regular work. All model endpoints, including the judge, were accessed under enterprise agreements with Vertex AI and Azure OpenAI, which exclude submitted prompts and outputs from provider model training. No data was retained beyond the terms

of these agreements. The Enterprise KB and OCR pipeline cannot be released because they are proprietary. We therefore include MMLongBench-Doc to enable independent reproduction of the main text-versus-image comparison.

References

- Anna Bavaresco, Raffaella Bernardi, Leonardo Bertolazzi, Desmond Elliott, Raquel Fernández, Albert Gatt, Esam Ghaleb, Mario Giulianelli, Michael Hanna, Alexander Koller, André F. T. Martins, Philipp Mondorf, Vera Neplenbroek, Sandro Pezzelle, Barbara Plank, David Schlangen, Alessandro Suglia, Aditya K. Surikuchi, Ece Takmaz, and Alberto Testoni. 2025. [LLMs instead of human judges? a large scale empirical study across 20 NLP evaluation tasks](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 238–255. Association for Computational Linguistics.
- David Berghaus, Armin Berger, Lars Hillebrand, Kostadin Cvejoski, and Rafet Sifa. 2025. [Multi-modal vision vs. text-based parsing: Benchmarking LLM strategies for invoice processing](#). In *2025 IEEE International Conference on Big Data*, pages 2519–2526. IEEE.
- Lukas Blecher, Guillem Cucurull, Thomas Scialom, and Robert Stojnic. 2024. [Nougat: Neural optical understanding for academic documents](#). In *International Conference on Learning Representations*.
- Jaemin Cho, Debanjan Mahata, Ozan Irsoy, Yujie He, and Mohit Bansal. 2024. [M3DocRAG: Multi-modal retrieval is what you need for multi-page multi-document understanding](#). *arXiv preprint arXiv:2411.04952*.
- Chao Deng, Jiale Yuan, Pi Bu, Peijie Wang, Zhong-Zhi Li, Jian Xu, Xiao-Hui Li, Yuan Gao, Jun Song, Bo Zheng, and Cheng-Lin Liu. 2025. [LongDocURL: a comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1135–1159. Association for Computational Linguistics.
- Kuicai Dong, Yujing Chang, Shijie Huang, Yasheng Wang, Ruiming Tang, and Yong Liu. 2025. [Benchmarking retrieval-augmented multimodal generation for document question answering](#). In *Advances in Neural Information Processing Systems*, volume 38.
- Manuel Faysse, Hugues Sibille, Tony Wu, Bilel Omrani, Gautier Viaud, Céline Hudelot, and Pierre Colombo. 2025. [ColPali: Efficient document retrieval with vision language models](#). In *International Conference on Learning Representations (ICLR)*. ArXiv:2407.01449.

- Yupan Huang, Tengchao Lv, Lei Cui, Yutong Lu, and Furu Wei. 2022. [LayoutLMv3: Pre-training for document AI with unified text and image masking](#). In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 4083–4091. Association for Computing Machinery.
- Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2023. [LLMLingua: Compressing prompts for accelerated inference of large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13358–13376. Association for Computational Linguistics.
- Kenton Lee, Mandar Joshi, Iulia Raluca Turc, Hexiang Hu, Fangyu Liu, Julian Martin Eisenschlos, Urvasi Khandelwal, Peter Shaw, Ming-Wei Chang, and Kristina Toutanova. 2023. [Pix2Struct: Screenshot parsing as pretraining for visual language understanding](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 18893–18912. PMLR.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive NLP tasks](#). In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Yucheng Li, Bo Dong, Frank Guerin, and Chenghua Lin. 2023. [Compressing context to enhance inference efficiency of large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6342–6353. Association for Computational Linguistics.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. [Lost in the middle: How language models use long contexts](#). *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. [G-eval: NLG evaluation using GPT-4 with better human alignment](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2511–2522. Association for Computational Linguistics.
- Xueguang Ma, Sheng-Chieh Lin, Minghan Li, Wenhu Chen, and Jimmy Lin. 2024a. [Unifying multimodal retrieval via document screenshot embedding](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6492–6505, Miami, Florida, USA. Association for Computational Linguistics.
- Yubo Ma, Yuhang Zang, Liangyu Chen, Meiqi Chen, Yizhu Jiao, Xinze Li, Xinyuan Lu, Ziyu Liu, Yan Ma, Xiaoyi Dong, Pan Zhang, Liangming Pan, Yugang Jiang, Jiaqi Wang, Yixin Cao, and Aixin Sun. 2024b. [Mmlongbench-doc: Benchmarking long-context document understanding with visualizations](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 95963–96010.
- Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. 2021. [DocVQA: A dataset for VQA on document images](#). In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2199–2208. IEEE.
- Ahmed Nassar, Andres Marafioti, Matteo Omenetti, Maksym Lysak, Nikolaos Livathinos, Christoph Auer, Lucas Morin, Rafael Teixeira de Lima, Yusik Kim, A. Said Gurbuz, Michele Dolfi, Miquel Farré, and Peter W. J. Staar. 2025. [SmolDocling: An ultra-compact vision-language model for end-to-end multimodal document conversion](#). In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21972–21983. IEEE.
- Jon Saad-Falcon, Joe Barrow, Alexa Siu, Ani Nenkova, Seunghyun Yoon, Ryan A. Rossi, and Franck Dernoncourt. 2024. [PDFTriage: Question answering over long, structured documents](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 153–169. Association for Computational Linguistics.
- Jiyuan Shen, Peiyue Yuan, Atin Ghosh, Yifan Mai, and Daniel Dahlmeier. 2026. [OCR or not? rethinking document information extraction in the MLLMs era with real-world large-scale datasets](#). In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 385–396. Association for Computational Linguistics.
- Ray Smith. 2007. [An overview of the Tesseract OCR engine](#). In *Ninth International Conference on Document Analysis and Recognition*, volume 2, pages 629–633. IEEE.
- Jordy Van Landeghem, Rubèn Tito, Łukasz Borchmann, Michał Pietruszka, Paweł Józiać, Rafał Powalski, Dawid Jurkiewicz, Mickaël Coustaty, Bertrand Ackaert, Ernest Valveny, Matthew B. Blaschko, Sien Moens, and Tomasz Stanisławek. 2023. [Document understanding dataset and evaluation \(DUDE\)](#). In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 19471–19483. IEEE.
- Shi Yu, Chaoyue Tang, Bokai Xu, Junbo Cui, Junhao Ran, Yukun Yan, Zhenghao Liu, Shuo Wang, Xu Han, Zhiyuan Liu, and Maosong Sun. 2025. [VisRAG: Vision-based retrieval-augmented generation on multi-modality documents](#). In *International Conference on Learning Representations (ICLR)*. ArXiv:2410.10594.
- Qintong Zhang, Victor Shea-Jay Huang, Bin Wang, Junyuan Zhang, Zhengren Wang, Hao Liang, Shawn Wang, Matthieu Lin, Conghui He, and Wentao Zhang.

Table 8: Direct OCR comparison against Tesseract. Scores report End2End recall under case-sensitive matching.

Dataset	End2End Recall (case-sensitive)	
	Tesseract	In-house
CORD-1k	0.474	0.881
DeepForm	0.668	0.970
FUNSD	0.738	0.815

2024. [Document parsing unveiled: Techniques, challenges, and prospects for structured information extraction](#). *arXiv preprint arXiv:2410.21169*.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhonghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. [Judging LLM-as-a-judge with MT-Bench and chatbot arena](#). In *Advances in Neural Information Processing Systems*, volume 36, pages 46595–46623.

A Experimental Settings and Provenance

Input and inference. Images use **300-DPI PDF page renders**. Text uses **page text**, a proprietary OCR pipeline outperforming Tesseract (Smith, 2007) (Table 8). All runs use the same QA prompt (Appendix E), no learned retriever, seed 30, a **3,000-token answer limit**, and **at most 50 page images** per request. Temperature is 0 for all four models.

Latency measurement. Provider-side automatic retries are disabled for all API calls. Reported latency is the wall-clock time of the single call that produced the answer, excluding any internal retry overhead the provider would otherwise add.

Judging. GPT-5.2 judges answers across **three runs** (8,000-token limit, two parallel calls per batch) (Zheng et al., 2023; Liu et al., 2023; Bavaresco et al., 2025). Questions are batched in fixed document order.

Model Identifiers. Run identifiers are `gemini-3.1-flash-lite`, `gemini-3.1-pro-preview`, `gpt-5.4`, and `gpt-5.4-mini`. These are the exact strings passed at inference time.

B Dataset Details

MMLongBench-Doc. MMLongBench-Doc contains 135 PDFs with 6,522 pages, averaging 48.3 pages per document (median 28, range 9–468) (Ma et al., 2024b). We use the current 1,091-question release. Questions include 485 single-page (44.5%),

360 cross-page (33.0%), and 246 unanswerable (22.5%). On answerable questions, the evidence labels are multi-label: Pure-text 298 (35.3%), Table 218 (25.8%), Chart 178 (21.1%), Image 293 (34.7%), and Layout 119 (14.1%). Answer formats include Integer 34.2%, String 29.5%, Float 18.9%, and List 17.3%. Questions cite 1.88 evidence pages on average (9.08 distinct evidence pages per document).

Enterprise KB (internal). The enterprise KB contains 1,447 documents (8,296 pages; avg. 5.7 pages/doc) with 5,169 human-reviewed questions (avg. 3.6 questions/doc). Answer mode: visual 3,314 (64.1%), cross-modal 1,527 (29.5%), textual 328 (6.3%). Difficulty: easy 3,731 (72.2%), medium 1,425 (27.6%), hard 13 (0.3%). 93.6% of questions require visual or cross-modal interpretation. Questions cite 1.05 evidence pages on average (2.42 distinct evidence pages per document).

C Full-corpus results

This appendix reports the per-model results underlying the main text over *all* documents, with the image budget still capped at 50 pages. Documents longer than 50 pages are therefore truncated under **IMAGE** and **TEXT+IMAGE**, while **TEXT** sees the full document. We report the effect of the ≤ 50 -page restriction on accuracy (Table 9) and latency (Table 10), the full length-bucket breakdown extending past the cap (Figure 3).

The restriction isolates the page-cap truncation effect. On the Enterprise KB, restricting to ≤ 50 pages changes every accuracy and latency cell by less than 0.001 (Tables 9, 10): only 6 of 5,169 questions come from longer documents, so the subset and the full corpus are effectively the same measurement. On MMLongBench-Doc, where 25% of documents exceed the cap, the restriction has a clear and localized effect. The accuracy gain from restricting concentrates entirely in **IMAGE/FULL**, where it reaches +0.041 to +0.058 across the four models, and is an order of magnitude smaller for **TEXT/FULL** and for every ORACLE cell. The latency change mirrors this: the large reductions (−1.0 to −1.6 s) fall on **IMAGE** and **TEXT+IMAGE** under **FULL**, while **TEXT** and all ORACLE cells barely move. Both patterns identify the same cause: only the image conditions on long documents are affected by the restriction, because only they are truncated by the page cap.

Table 9: Effect of restricting to the ≤ 50 -page subset. Accuracy is mean $\pm$ std across runs; $\Delta = \text{acc}_{\leq 50} - \text{acc}_{\text{all}}$. **Bold** Δ marks $|\Delta| \geq 0.02$. Large positive Δ concentrates in **IMAGE**/FULL, reflecting vision-input truncation on documents longer than the page cap.

Model	Representation	Regime	Acc $_{\leq 50}$	Acc $_{\text{all}}$	Δ
Enterprise KB					
GPT-5.4	TEXT	FULL	0.6634 $\pm$ 0.0059	0.6626 $\pm$ 0.0059	+0.0008
GPT-5.4	TEXT	ORACLE	0.6387 $\pm$ 0.0033	0.6382 $\pm$ 0.0033	+0.0005
GPT-5.4	IMAGE	FULL	0.6792 $\pm$ 0.0043	0.6788 $\pm$ 0.0043	+0.0004
GPT-5.4	IMAGE	ORACLE	0.6774 $\pm$ 0.0018	0.6769 $\pm$ 0.0018	+0.0005
GPT-5.4	TEXT+IMAGE	FULL	0.7344 $\pm$ 0.0051	0.7338 $\pm$ 0.0052	+0.0006
GPT-5.4	TEXT+IMAGE	ORACLE	0.7329 $\pm$ 0.0039	0.7324 $\pm$ 0.0039	+0.0005
GPT-5.4-mini	TEXT	FULL	0.6277 $\pm$ 0.0033	0.6269 $\pm$ 0.0033	+0.0007
GPT-5.4-mini	TEXT	ORACLE	0.6122 $\pm$ 0.0015	0.6116 $\pm$ 0.0015	+0.0006
GPT-5.4-mini	IMAGE	FULL	0.6720 $\pm$ 0.0027	0.6714 $\pm$ 0.0027	+0.0006
GPT-5.4-mini	IMAGE	ORACLE	0.6825 $\pm$ 0.0027	0.6821 $\pm$ 0.0028	+0.0005
GPT-5.4-mini	TEXT+IMAGE	FULL	0.7148 $\pm$ 0.0054	0.7142 $\pm$ 0.0053	+0.0006
GPT-5.4-mini	TEXT+IMAGE	ORACLE	0.7309 $\pm$ 0.0020	0.7304 $\pm$ 0.0020	+0.0005
Gemini 3.1 Pro	TEXT	FULL	0.6335 $\pm$ 0.0031	0.6330 $\pm$ 0.0031	+0.0005
Gemini 3.1 Pro	TEXT	ORACLE	0.6179 $\pm$ 0.0029	0.6174 $\pm$ 0.0028	+0.0004
Gemini 3.1 Pro	IMAGE	FULL	0.7280 $\pm$ 0.0016	0.7276 $\pm$ 0.0016	+0.0005
Gemini 3.1 Pro	IMAGE	ORACLE	0.7409 $\pm$ 0.0035	0.7406 $\pm$ 0.0035	+0.0003
Gemini 3.1 Pro	TEXT+IMAGE	FULL	0.7275 $\pm$ 0.0034	0.7271 $\pm$ 0.0035	+0.0005
Gemini 3.1 Pro	TEXT+IMAGE	ORACLE	0.7364 $\pm$ 0.0017	0.7361 $\pm$ 0.0017	+0.0003
Gemini 3.1 Flash-Lite	TEXT	FULL	0.5942 $\pm$ 0.0009	0.5938 $\pm$ 0.0008	+0.0004
Gemini 3.1 Flash-Lite	TEXT	ORACLE	0.5762 $\pm$ 0.0024	0.5757 $\pm$ 0.0024	+0.0005
Gemini 3.1 Flash-Lite	IMAGE	FULL	0.7112 $\pm$ 0.0025	0.7110 $\pm$ 0.0025	+0.0002
Gemini 3.1 Flash-Lite	IMAGE	ORACLE	0.7294 $\pm$ 0.0015	0.7292 $\pm$ 0.0015	+0.0003
Gemini 3.1 Flash-Lite	TEXT+IMAGE	FULL	0.7141 $\pm$ 0.0002	0.7138 $\pm$ 0.0002	+0.0004
Gemini 3.1 Flash-Lite	TEXT+IMAGE	ORACLE	0.7243 $\pm$ 0.0010	0.7241 $\pm$ 0.0010	+0.0003
MMLongBench-Doc					
GPT-5.4	TEXT	FULL	0.5990 $\pm$ 0.0093	0.5927 $\pm$ 0.0086	+0.0063
GPT-5.4	TEXT	ORACLE	0.5589 $\pm$ 0.0065	0.5615 $\pm$ 0.0077	-0.0026
GPT-5.4	IMAGE	FULL	0.6857 $\pm$ 0.0078	0.6348 $\pm$ 0.0082	+0.0509
GPT-5.4	IMAGE	ORACLE	0.6672 $\pm$ 0.0043	0.6585 $\pm$ 0.0039	+0.0087
GPT-5.4	TEXT+IMAGE	FULL	0.6830 $\pm$ 0.0080	0.6636 $\pm$ 0.0079	+0.0193
GPT-5.4	TEXT+IMAGE	ORACLE	0.6734 $\pm$ 0.0027	0.6643 $\pm$ 0.0027	+0.0091
GPT-5.4-mini	TEXT	FULL	0.5466 $\pm$ 0.0046	0.5430 $\pm$ 0.0038	+0.0036
GPT-5.4-mini	TEXT	ORACLE	0.5268 $\pm$ 0.0093	0.5340 $\pm$ 0.0097	-0.0072
GPT-5.4-mini	IMAGE	FULL	0.6333 $\pm$ 0.0038	0.5923 $\pm$ 0.0025	+0.0410
GPT-5.4-mini	IMAGE	ORACLE	0.6343 $\pm$ 0.0052	0.6279 $\pm$ 0.0025	+0.0065
GPT-5.4-mini	TEXT+IMAGE	FULL	0.6424 $\pm$ 0.0073	0.6236 $\pm$ 0.0040	+0.0187
GPT-5.4-mini	TEXT+IMAGE	ORACLE	0.6426 $\pm$ 0.0065	0.6354 $\pm$ 0.0054	+0.0072
Gemini 3.1 Pro	TEXT	FULL	0.6276 $\pm$ 0.0072	0.6053 $\pm$ 0.0084	+0.0223
Gemini 3.1 Pro	TEXT	ORACLE	0.6413 $\pm$ 0.0074	0.6320 $\pm$ 0.0066	+0.0093
Gemini 3.1 Pro	IMAGE	FULL	0.6902 $\pm$ 0.0040	0.6326 $\pm$ 0.0050	+0.0576
Gemini 3.1 Pro	IMAGE	ORACLE	0.7113 $\pm$ 0.0076	0.7039 $\pm$ 0.0063	+0.0073
Gemini 3.1 Pro	TEXT+IMAGE	FULL	0.6233 $\pm$ 0.0060	0.6077 $\pm$ 0.0042	+0.0156
Gemini 3.1 Pro	TEXT+IMAGE	ORACLE	0.6912 $\pm$ 0.0030	0.6849 $\pm$ 0.0013	+0.0064
Gemini 3.1 Flash-Lite	TEXT	FULL	0.5109 $\pm$ 0.0006	0.5108 $\pm$ 0.0004	+0.0000
Gemini 3.1 Flash-Lite	TEXT	ORACLE	0.5292 $\pm$ 0.0006	0.5228 $\pm$ 0.0004	+0.0065
Gemini 3.1 Flash-Lite	IMAGE	FULL	0.6145 $\pm$ 0.0093	0.5778 $\pm$ 0.0074	+0.0367
Gemini 3.1 Flash-Lite	IMAGE	ORACLE	0.6253 $\pm$ 0.0027	0.6205 $\pm$ 0.0033	+0.0048
Gemini 3.1 Flash-Lite	TEXT+IMAGE	FULL	0.6199 $\pm$ 0.0056	0.6101 $\pm$ 0.0011	+0.0097
Gemini 3.1 Flash-Lite	TEXT+IMAGE	ORACLE	0.6128 $\pm$ 0.0018	0.6101 $\pm$ 0.0017	+0.0026

The text advantage appears only past the cap.

Figure 3 extends the length-bucket view to the full corpus (shaded region marks buckets above the 50-page cap). Within the un-truncated range, image-bearing inputs lead at every length, matching the main text. The crossover in favor of **TEXT** emerges only in the truncated buckets: on MMLongBench-Doc, **IMAGE** accuracy falls to $\sim 42\%$ at 100–150 pages and $\sim 39\%$ beyond 150, while **TEXT** holds near 58%. This is the behavior that motivates reporting the ≤ 50 -page subset in the main text: the crossover is a property of the fixed image budget, not of the modalities, and it disappears once every representation sees the complete page set.

D Error Category Analysis

Tables 11 and 12 break down errors by model, representation, and regime (tagged by judge; Appendix E). Categories are non-exclusive.

On the KB, `partial_correctness` is the most frequent reported category (44–57%), `extra_incorrect_information` follows (18–25%), and switching from ORACLE to FULL barely changes the profile. On MMLongBench-Doc, `answered_when_unanswerable` is high (36–53%), consistent with the fact that 22.5% of questions are unanswerable by design; this category means the model *gave* an answer to

Table 10: Effect of the ≤ 50 -page restriction on latency (seconds/question, mean $\pm$ std); $\Delta = \text{lat}_{\leq 50} - \text{lat}_{\text{all}}$ (negative = subset is faster). **Bold** Δ marks $|\Delta| \geq 0.5\text{s}$. Large negative Δ concentrates in **IMAGE**/**TEXT+IMAGE** under FULL: the excluded long documents are the ones that inflate vision latency.

Model	Representation	Regime	Lat _{≤ 50}	Lat _{all}	Δ (s)
Enterprise KB					
GPT-5.4	TEXT	FULL	3.42 $\pm$ 0.18	3.42 $\pm$ 0.18	-0.003
GPT-5.4	TEXT	ORACLE	3.46 $\pm$ 0.29	3.46 $\pm$ 0.29	-0.001
GPT-5.4	IMAGE	FULL	5.43 $\pm$ 1.07	5.44 $\pm$ 1.08	-0.011
GPT-5.4	IMAGE	ORACLE	4.97 $\pm$ 0.42	4.98 $\pm$ 0.42	-0.009
GPT-5.4	TEXT+IMAGE	FULL	5.71 $\pm$ 0.43	5.73 $\pm$ 0.43	-0.013
GPT-5.4	TEXT+IMAGE	ORACLE	4.48 $\pm$ 0.42	4.49 $\pm$ 0.42	-0.004
GPT-5.4-mini	TEXT	FULL	1.78 $\pm$ 0.42	1.78 $\pm$ 0.42	-0.000
GPT-5.4-mini	TEXT	ORACLE	1.68 $\pm$ 0.20	1.68 $\pm$ 0.20	-0.001
GPT-5.4-mini	IMAGE	FULL	3.36 $\pm$ 0.13	3.37 $\pm$ 0.13	-0.011
GPT-5.4-mini	IMAGE	ORACLE	2.55 $\pm$ 0.15	2.55 $\pm$ 0.15	-0.002
GPT-5.4-mini	TEXT+IMAGE	FULL	3.26 $\pm$ 0.16	3.27 $\pm$ 0.16	-0.012
GPT-5.4-mini	TEXT+IMAGE	ORACLE	2.42 $\pm$ 0.15	2.43 $\pm$ 0.15	-0.000
Gemini 3.1 Pro	TEXT	FULL	6.21 $\pm$ 0.25	6.21 $\pm$ 0.25	-0.001
Gemini 3.1 Pro	TEXT	ORACLE	5.57 $\pm$ 0.25	5.57 $\pm$ 0.25	-0.001
Gemini 3.1 Pro	IMAGE	FULL	5.68 $\pm$ 0.15	5.68 $\pm$ 0.15	-0.005
Gemini 3.1 Pro	IMAGE	ORACLE	5.69 $\pm$ 0.36	5.69 $\pm$ 0.36	+0.000
Gemini 3.1 Pro	TEXT+IMAGE	FULL	5.81 $\pm$ 0.22	5.82 $\pm$ 0.22	-0.009
Gemini 3.1 Pro	TEXT+IMAGE	ORACLE	5.36 $\pm$ 0.37	5.36 $\pm$ 0.37	+0.000
Gemini 3.1 Flash-Lite	TEXT	FULL	1.40 $\pm$ 0.01	1.40 $\pm$ 0.01	-0.001
Gemini 3.1 Flash-Lite	TEXT	ORACLE	1.29 $\pm$ 0.00	1.29 $\pm$ 0.00	-0.000
Gemini 3.1 Flash-Lite	IMAGE	FULL	3.26 $\pm$ 0.60	3.27 $\pm$ 0.60	-0.008
Gemini 3.1 Flash-Lite	IMAGE	ORACLE	2.42 $\pm$ 0.38	2.42 $\pm$ 0.38	+0.000
Gemini 3.1 Flash-Lite	TEXT+IMAGE	FULL	3.20 $\pm$ 0.32	3.21 $\pm$ 0.32	-0.007
Gemini 3.1 Flash-Lite	TEXT+IMAGE	ORACLE	2.52 $\pm$ 0.31	2.52 $\pm$ 0.31	+0.000
MMLongBench-Doc					
GPT-5.4	TEXT	FULL	4.78 $\pm$ 0.24	4.99 $\pm$ 0.27	-0.207
GPT-5.4	TEXT	ORACLE	4.21 $\pm$ 0.32	4.15 $\pm$ 0.30	+0.067
GPT-5.4	IMAGE	FULL	13.37 $\pm$ 0.61	14.75 $\pm$ 0.65	-1.386
GPT-5.4	IMAGE	ORACLE	8.03 $\pm$ 0.75	7.99 $\pm$ 0.74	+0.044
GPT-5.4	TEXT+IMAGE	FULL	13.36 $\pm$ 0.51	14.92 $\pm$ 0.52	-1.555
GPT-5.4	TEXT+IMAGE	ORACLE	7.83 $\pm$ 0.67	7.83 $\pm$ 0.65	-0.004
GPT-5.4-mini	TEXT	FULL	2.22 $\pm$ 0.27	2.32 $\pm$ 0.30	-0.103
GPT-5.4-mini	TEXT	ORACLE	2.00 $\pm$ 0.23	1.99 $\pm$ 0.23	+0.013
GPT-5.4-mini	IMAGE	FULL	10.20 $\pm$ 0.31	11.53 $\pm$ 0.33	-1.332
GPT-5.4-mini	IMAGE	ORACLE	5.05 $\pm$ 0.36	5.07 $\pm$ 0.37	-0.020
GPT-5.4-mini	TEXT+IMAGE	FULL	10.09 $\pm$ 0.36	11.50 $\pm$ 0.36	-1.405
GPT-5.4-mini	TEXT+IMAGE	ORACLE	5.03 $\pm$ 0.31	5.04 $\pm$ 0.31	-0.014
Gemini 3.1 Pro	TEXT	FULL	7.57 $\pm$ 0.31	7.82 $\pm$ 0.48	-0.246
Gemini 3.1 Pro	TEXT	ORACLE	6.10 $\pm$ 0.22	6.12 $\pm$ 0.44	-0.015
Gemini 3.1 Pro	IMAGE	FULL	8.40 $\pm$ 0.50	9.45 $\pm$ 1.17	-1.046
Gemini 3.1 Pro	IMAGE	ORACLE	6.88 $\pm$ 0.69	7.06 $\pm$ 0.65	-0.180
Gemini 3.1 Pro	TEXT+IMAGE	FULL	10.12 $\pm$ 2.31	11.61 $\pm$ 2.86	-1.492
Gemini 3.1 Pro	TEXT+IMAGE	ORACLE	6.80 $\pm$ 0.72	7.01 $\pm$ 0.81	-0.211
Gemini 3.1 Flash-Lite	TEXT	FULL	2.10 $\pm$ 0.06	2.31 $\pm$ 0.08	-0.209
Gemini 3.1 Flash-Lite	TEXT	ORACLE	1.72 $\pm$ 0.09	1.74 $\pm$ 0.08	-0.022
Gemini 3.1 Flash-Lite	IMAGE	FULL	5.88 $\pm$ 0.52	6.58 $\pm$ 0.57	-0.694
Gemini 3.1 Flash-Lite	IMAGE	ORACLE	3.59 $\pm$ 0.44	3.74 $\pm$ 0.41	-0.152
Gemini 3.1 Flash-Lite	TEXT+IMAGE	FULL	6.93 $\pm$ 0.28	8.05 $\pm$ 0.20	-1.119
Gemini 3.1 Flash-Lite	TEXT+IMAGE	ORACLE	4.06 $\pm$ 0.16	4.19 $\pm$ 0.13	-0.124

an unanswerable question (over-answering), and its rate is largely stable across regimes rather than growing with the ORACLE $\rightarrow$ FULL shift. numerical_imprecision is also frequent (25–41%); Integer and Float are common answer formats in this benchmark.

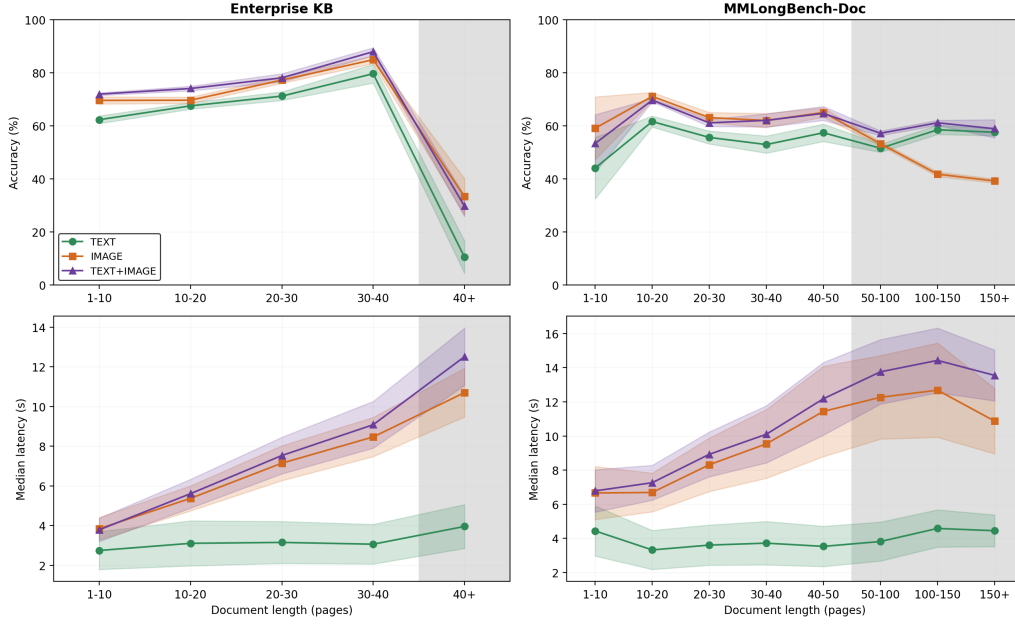


Figure 3: Full-corpus accuracy (top) and median latency (bottom) versus document-length bucket under FULL retrieval, aggregated across all four models. The shaded region marks buckets above the 50-page image cap, where **IMAGE** and **TEXT+IMAGE** inputs are truncated while **TEXT** sees the full document. Within the un-truncated range, image-bearing inputs lead on accuracy at every length; the crossover in favor of **TEXT** appears only in the truncated buckets, and image latency falls there as truncation caps the number of page images processed. Bands show $\pm$ std across models.

Table 11: Non-exclusive error-category shares (% of incorrect answers) by model, representation, and regime on Enterprise KB. Left block: ORACLE (gold evidence pages); right block: FULL (full document). Columns need not sum to 100%.

Model	ORACLE			FULL		
	TEXT	IMAGE	TEXT+IMAGE	TEXT	IMAGE	TEXT+IMAGE
<i>partial_correctness</i>						
Gemini 3.1 Flash-Lite	44.9	53.0	52.9	46.0	52.0	52.1
Gemini 3.1 Pro	50.5	56.8	56.0	49.5	52.0	52.0
GPT-5.4	51.5	49.1	54.7	52.4	45.5	52.0
GPT-5.4-mini	49.4	48.1	53.7	51.0	49.3	51.5
<i>extra_incorrect_information</i>						
Gemini 3.1 Flash-Lite	18.7	18.6	19.8	21.2	23.3	25.5
Gemini 3.1 Pro	18.1	17.7	20.4	20.4	22.9	22.4
GPT-5.4	20.8	19.9	22.4	22.6	21.1	23.7
GPT-5.4-mini	21.8	18.5	21.6	23.8	20.8	24.4
<i>contradiction</i>						
Gemini 3.1 Flash-Lite	19.1	18.3	17.1	19.2	19.6	21.4
Gemini 3.1 Pro	16.3	16.9	15.8	19.9	19.2	19.9
GPT-5.4	21.9	23.1	19.0	20.6	23.0	20.2
GPT-5.4-mini	21.7	21.3	19.5	21.9	21.1	20.0
<i>answered_when_unanswerable</i>						
Gemini 3.1 Flash-Lite	20.7	10.4	10.8	17.7	6.6	4.9
Gemini 3.1 Pro	14.8	8.3	9.8	15.4	10.6	9.3
GPT-5.4	9.3	8.3	6.1	9.6	9.1	7.2
GPT-5.4-mini	9.6	7.3	4.7	9.3	6.6	4.9
<i>numerical_imprecision</i>						
Gemini 3.1 Flash-Lite	6.0	4.6	5.3	5.9	5.6	5.9
Gemini 3.1 Pro	4.8	4.0	4.1	4.4	4.0	4.2
GPT-5.4	6.9	10.3	6.8	6.2	10.3	5.7
GPT-5.4-mini	8.4	11.7	7.9	7.1	10.6	7.5

Table 12: Non-exclusive error-category shares (% of incorrect answers) by model, representation, and regime on MMLongBench-Doc. Left block: ORACLE (gold evidence pages); right block: FULL (full document). Columns need not sum to 100%.

Model	ORACLE			FULL		
	TEXT	IMAGE	TEXT+IMAGE	TEXT	IMAGE	TEXT+IMAGE
<i>partial_correctness</i>						
Gemini 3.1 Flash-Lite	9.6	11.7	11.7	10.1	10.8	10.6
Gemini 3.1 Pro	9.3	11.7	10.5	10.8	8.5	10.6
GPT-5.4	12.3	10.4	10.2	8.7	7.6	9.1
GPT-5.4-mini	9.0	9.2	8.8	11.0	8.8	9.3
<i>extra_incorrect_information</i>						
Gemini 3.1 Flash-Lite	4.0	5.5	6.5	6.0	5.7	8.2
Gemini 3.1 Pro	4.4	5.7	4.4	8.2	6.7	7.8
GPT-5.4	6.1	5.3	5.5	5.8	5.1	5.5
GPT-5.4-mini	5.8	4.0	5.9	5.4	7.0	6.6
<i>contradiction</i>						
Gemini 3.1 Flash-Lite	6.7	8.6	8.4	14.0	8.9	8.9
Gemini 3.1 Pro	8.8	9.2	7.0	8.7	7.5	7.1
GPT-5.4	9.0	8.6	9.1	11.4	9.8	10.5
GPT-5.4-mini	10.8	8.2	9.5	12.2	11.1	9.5
<i>answered_when_unanswerable</i>						
Gemini 3.1 Flash-Lite	51.6	40.4	39.0	36.3	38.2	36.5
Gemini 3.1 Pro	52.9	42.9	45.3	52.7	44.3	42.4
GPT-5.4	43.6	46.8	44.6	40.5	47.2	39.7
GPT-5.4-mini	44.0	45.5	42.8	37.1	41.4	37.7
<i>numerical_imprecision</i>						
Gemini 3.1 Flash-Lite	30.5	35.6	37.4	37.6	39.3	40.7
Gemini 3.1 Pro	26.5	32.1	33.6	24.9	35.8	34.8
GPT-5.4	32.4	31.6	33.1	36.7	32.1	36.1
GPT-5.4-mini	33.7	33.9	34.7	38.6	35.3	40.6

E Prompt Templates

E.1 QA Prompt

All conditions use the prompt below; **IMAGE** and **TEXT+IMAGE** additionally append the rendered page images after the text context block.

QA Prompt Template

You are an expert question answering assistant. Your task is to provide a concise and accurate answer to the user's question, strictly based on the provided context.

Instructions:

1. Read the question and the provided context carefully.
2. In the "explanation" field, think step-by-step to determine how the context answers the question.
3. Formulate an answer that directly addresses the question.
4. Your answer must be derived solely from the information present in the context. Do not use outside knowledge or invent facts.
5. If the context does not contain enough information, state "Not Answerable".
6. Keep the answer as short as possible: a single value, phrase, or list of items.

Output Format:

```
{
  "explanation": "<reasoning>",
  "answer": "<concise answer>"
}
```

Question: {{question}}

Context:

E.2 LLM-as-Judge Prompt

The judge receives batches of question-answer pairs through the `{{qa_pairs}}` placeholder, following reference-guided evaluation protocols for open-ended model outputs (Zheng et al., 2023; Liu et al., 2023; Bavaresco et al., 2025).

LLM-as-Judge Prompt Template

You are an expert evaluator assessing answers in a Question and Answer system. For each question-answer pair you receive:

1. A unique question ID
2. The question text
3. The predicted answer
4. The ground truth answer

Determine:

1. Whether the predicted answer is semantically equivalent to the ground truth.
2. All applicable error or warning categories.

Evaluation Instructions:

- Evaluate based on semantic meaning, not exact wording.
- Be flexible with phrasing, terminology, units, abbreviations, and formatting.
- Mark incorrect only if: (1) core meaning differs, (2) contains factually incorrect information, (3) contradicts the ground truth, or (4) omits key information.
- For numeric values: accept more-precise expansions of rounded ground truths; penalize rounding that loses accuracy.
- For lists: every item must match; order does not matter.
- For multiple-choice: accept the option letter, text, or both.

ERROR CATEGORIES (incorrect answers only):

partial_correctness,
extra_incorrect_information, contradiction,

```
    overly_vague, incorrect_yes_no,
    answered_when_unanswerable,
    numerical_imprecision, minor_spelling_error
WARNING CATEGORIES (correct answers only):
    ground_truth_rounding,
    ground_truth_spelling_error,
    redundant_information, extra_information,
    missing_information, other
Output Format:
{ "evaluations": [
  { "question_id": "<id>",
    "is_correct": true/false,
    "explanation": "<pred> vs <truth>",
    "errors": [{"class": "...",
                  "explanation": "..."}],
    "warnings": [{"class": "...",
                   "explanation": "..."}]
  }, ...
] }
```